

Generally, Pluto is considered the farthest planet from the Sun. As its orbit lies inside the orbit of Neptune, sometimes Neptune becomes the farthest planet from the Sun. Neptune was farthest from 1979 to 1999 but from 11th February, 1999. Pluto will be the farthest planet for the next 228 years. Hence, it can be seen that sometimes Pluto and sometimes Neptune become the farthest planet from the Sun. As in 2006, Pluto was declared a dwarf planet, hence now Neptune is the correct answer for this question.

9. Although Pluto is the farthest planet, its orbital path is excessively elliptical and crosses other planets orbital path, resulting the period starting from 1979 for 20 years Pluto is not the farthest planet, which planet is farthest planet ?

(a) Jupiter (b) Saturn
(c) Uranus (d) Neptune

I.A.S. (Pre) 1993

Ans. (d)

See the explanation of the above question.

10. Nix and Hydra are the moons of –

(a) Pluto (b) Venus
(c) Jupiter (d) Mercury

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (a)

Pluto is a dwarf planet of the Solar System, it has five satellites (moons). According to the distance from Pluto (from inner to outer side) their names are Charon, Styx, Nix, Kerberos and Hydra. Charon is the biggest satellite. There are 95 known satellites of Jupiter and till now there is no information about the existence of any satellite of Venus and Mercury.

xi. The Moon

*The moon is an astronomical body that orbits planet Earth and is Earth's only permanent natural satellite. The diameter of the Moon is 3,475 km. *The minimum and maximum distance between the Earth and Moon is **363300** km and **405500** km respectively. *The Moon makes a complete orbit around Earth once every **27** days, **7** hours and **43** minutes. It is the same time that is taken by the Moon to complete its one rotation. The moon is tilted on its axis, due to which its appearance (size) changes with its distance from the Earth. Because of this effect, about 59% of the Moon's surface can actually be seen from Earth. *Sound requires a medium to

travel. Since there is **no atmosphere** on the Moon, When two astronomers talk to each other it is not audible. ***Solar eclipses** occur at the new moon, when the moon comes in between the sun and the earth. In contrast, **lunar eclipses** occur at full moon, when the earth is in between the sun and the moon. The type and duration of a lunar eclipse depend upon the orbital tilt and alignment of the moon. *When the moon is closest to Earth (363300 km), then this condition is called **perigee**. It is during this time that **high tides** occur, because the gravitational pull of the moon is at its maximum. When the moon is at its farthest distance from the earth, the condition is called **apogee**. This is the time when low tides occur because the gravitational pull of the Moon is minimum. *The low tides are **20%** smaller than the high tides. *A lunar perigee occurred on December 22nd, 1999 due to which the moon appeared to be the brightest. Again on November 14th, 2016 a full perigee occurred due to which the moon for the first time in over 60 years appeared to be the largest and the brightest. In common language, this phenomenon is known as **supermoon**. A supermoon occurs when the full moon coincides with the moon's orbit being closest (perigee) to Earth. *During a supermoon, the moon appears roughly 14%, larger in diameter and 30% brighter. *During a total lunar eclipse, earth completely blocks direct sunlight from reaching the moon. The only light reflected from the lunar surface has been refracted by Earth's Atmosphere. Due to the reddish colour, a totally eclipsed moon is sometimes called a **blood moon**. *A **Blue Moon** is an additional full moon that appears in a subdivision of a year-either as the third or fourth full moon in a season or a second full moon in a month of the common calendar. One lunar month is around **29.5 days** long. If we multiply 29.5 days by 12 months we get 354 days, which means there could be 12 full moons in a year. Since 11 days are still left in a year, approximately every 2.7154 years the condition for a Blue moon (one extra full moon) comes into existence. In every 2.7154 years there would be a month that has two full moons, however the difference between these two full moons will always be **29.5** days. Traditionally the **second full moon of the month** is called a Blue Moon. *Accordingly in the year 2009 there were two full moons in December, one on 2nd December and one on 31st December. The one that occurred on 31st December was called a Blue Moon. *According to the new definition of the Almanac the Blue Moon phenomenon is the occurrence of one extra full moon in an astronomical season. There are four astronomical seasons in a year- starting from 21st March to 23rd September (the Equinoxes) and the other two starting from 21st June to

22nd December (summer and winter solstice respectively).

Hence the astronomical seasons are like this –

1st Astronomical season (21st March to 20th June)

2nd Astronomical season (21st June to 22nd September)

3rd Astronomical season (23rd September to 21st December)

4th Astronomical season (22nd December to 20th March)

Every season is divided into 3 months. Hence every season will normally have 3 full moons, however in every 2.7154 years, one of the above 4 seasons will have 4 full moons instead of 3. The astronomical season that has the 4th full moon will be known as the season with the Blue Moon. It is to be noted however that the **3rd full moon will be considered as the Blue moon** and not the 4th one. It is because only then will the names of the other full moons, such as the moon before the solstice and the Moon after the solstice, fall at the proper times relative to the solstices and equinoxes. Hence, according to the above definition, the year 2010 witnessed 4 full moons the astronomical season starting from 23rd September to 22nd December. 25th September, 24th October, 21st November and 20th December witnessed full moons. The full moon of 21st November 2010 was referred to as a Blue Moon. *The weight of any body on a planet or satellite depends upon the mass of the body and the gravitational acceleration of the planet. (Weight = mass × gravitational acceleration). The gravitational acceleration on the moon is 1/6th of that of planet Earth. Hence, any body or object taken to the moon has its weight reduced to 1/6th of its weight on planet Earth. The area of the large black patch as seen from the Earth is called Lunar Mare or **Sea of Tranquility**. It was the landing site for the first crewed landing on the Moon on July 20, 1969. Astronauts **Neil Armstrong** and **Edwin Buzz Aldrin Junior** made a smooth touchdown in the Apollo 11 Lunar Module named Eagle. The third astronaut on this mission was **Michael Collins**, however, he did not step on the Moon. *The Apollo 11 returned to Earth and splashed down in the Pacific Ocean on July 24, 1969. The mission brought back substances from the surface of the Moon which weighed around 47 pounds. It was used for research purposes. ***Chandrayaan-1** was India's first lunar probe. It was launched by the Indian Space Research Organisation on 22 October, 2008 and operated until August 2009. India launched the spacecraft using a PSLV-XL rocket (PSLV-C11). The vehicle was inserted into lunar orbit on 8 November 2008. *The Indian Space Research Organisation (ISRO) set up a Deep Space Network (DSN) centre in Byalalu (Bengaluru)

to track Chandrayaan-1. ***Chandrayaan-2** is India's second lunar exploration mission after Chandrayaan-1 developed by the Indian Space Research Organisation, the mission was launched to the Moon by a Geosynchronous Satellite Launch Vehicle Mk III - M1 (GSLV-MkIII-M1) on July 22, 2019.

Chandrayaan-3 mission took off on July 14, 2023 from the Satish Dhawan Space Centre in Andhra Pradesh. On August 5, 2023 the mission entered lunar orbit. On August 17, the lander module separated from the propulsion module. The first and second deboosting occurred on August 18 and 20 respectively. On August 23, 2023 the lander landed on the south pole of the Moon. India's third lunar mission, Chandrayaan - 3 made history in August, 2023 by successfully achieving a soft landing on the south pole of the Moon.

With this, India became the first nation to land on the Moon's south pole and the fourth (after Russia, U.S.A. and China) to land successfully on Earth's only natural satellite.

1. When did man first land on the Moon?

- | | |
|----------|----------|
| (a) 1953 | (b) 1963 |
| (c) 1971 | (d) 1969 |

M.P.P.C.S. (Pre) 1995

Ans. (d)

On 20th July 1969 for the first time, man landed on the Moon. Three astronauts sent with the Apollo-11 mission to the Moon were Neil Armstrong, Michael Collins and Edwin Buzz Aldrin Junior. Among these astronauts, only Neil Armstrong and Edwin Aldrin walked on the surface of the Moon. On 16 July 1969, initiating its flight from Cape Kennedy Space Center, Florida (U.S.A.). Apollo-11 landed on the 'Sea of Tranquility' upon the Moon on 20 July 1969. On 21 July 1969 it started its return journey from the Moon towards the Earth and finally landed in the Pacific Ocean on 24 July, 1969.

2. The "Sea of Tranquility" lies on -

- | | |
|-------------|----------|
| (a) Earth | (b) Sun |
| (c) Jupiter | (d) Moon |

M.P.P.C.S. (Pre) 1995

Ans. (d)

The Area of large the black patch as seen from the Earth is called Lunar Mare or the "Sea of Tranquility" which is found on the large area of the Moon where the spacecraft Apollo-11 landed in 1969. Neil Armstrong was the first person to touch the surface of the Moon, followed by Edwin Aldrin Junior.

3. **Assertion (A) : The same face of the Moon always faces the Earth.**

Reason (R) : The Moon completes one rotation around its own axis in 23 and half days which is about the same time that it takes to orbit the Earth.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2005

Ans. (c)

The orbit of the Moon around the Earth is elliptical and tilted and therefore we can see only 59% of the Moon's surface at a time. The Moon takes 27 days, 7 hours and 43 minutes to rotate on its axis. This is also the time of one revolution of the Moon around the Earth. This is the reason why we always see the same face of the Moon. Therefore, statement A is correct but reason R is incorrect.

4. **Consider the following statements and answer the question with the help of codes given below :**

Assertion (A) : Only one side of the moon's surface is visible from the earth.

Reason(R) : The period of rotation of the moon on its axis is equal to the period of its revolution around the earth.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2006

Ans. (a)

We see only one side of the Moon's surface from Earth as the Moon's orbit around the Earth is elliptical and tilted. Earth's satellite the Moon rotates on its axis and revolves around the Earth in the same time of 27 days, 7 hrs and 43 minutes. So only one side of the Moon's surface is visible from the Earth, therefore statement 'A' is correct and reason 'R' is also correct. Therefore reason (R) explains statement (A) correctly.

5. **On the surface of the moon, two persons cannot listen to each other as –**

- (a) Their ears stops working at the Moon.
- (b) There is no atmosphere on the Moon.

- (c) They wear special types of space suits on the Moon.
- (d) Sound travels very slow on the Moon.

U.P.P.C.S. (Pre) 2003

Ans. (b)

For communication of sound waves, a material medium is needed and the atmosphere is such a medium. At the Moon the atmosphere is absent. That is why two people cannot listen to each other on the Moon.

6. **When does a Lunar eclipse take place?**

- (a) When the moon comes between the sun and the earth.
- (b) When the earth comes between the sun and the moon.
- (c) When the sun comes between the earth and the moon.
- (d) None of these.

M.P.P.C.S. (Pre) 1994

Ans. (b)

When the Earth comes between the Moon and the Sun, the Moon gets into the dark due to the obstruction of light rays coming from the Sun by the Earth. The type of the lunar eclipse and its length depends upon the relative position of the Moon and its orbital path. A lunar eclipse only occurs on the full moon day.

7. **Lunar Eclipse is caused when –**

- (a) Moon comes in between Earth and Sun
- (b) Earth comes between Sun and Moon
- (c) Sun comes between Earth and Moon
- (d) Sun and Moon coincide

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

See the explanation of the above question.

8. **In which situation lunar eclipse occurs ?**

- (a) Half Moon
- (b) New Moon
- (c) Full Moon
- (d) None of these

M.P.P.C.S. (Pre) 1992

Ans. (c)

See the explanation of the above question.

9. **The main condition responsible for greater brightness of the moon on the full moon day of 22nd Dec. 99 was :**

- (a) Aphelion
- (b) Perihelion
- (c) Apogee
- (d) Perigee

R.A.S./R.T.S.(Pre) 1999

Ans. (d)

The main condition responsible for the greater brightness of the moon on the full moon day of 22nd December, 1999 was perigee. Perigee is the closest point to the Earth.

10. **When the moon size is half of the full moon, the angle between the Sun, the Earth and the Moon is –**

- (a) 45°
- (b) 90°

(c) 180°

(d) 270°

U.P.P.C.S. (Pre) 2005

Ans. (b)

When the Moon size is half of the full Moon, the angle between the Sun, the Earth and the Moon is 90°.

11. Choose the odd one –

(a) Mars

(b) Mercury

(c) Moon

(d) Earth

Chhattisgarh P.C.S. (Pre) 2005

Ans. (c)

Mars, Mercury and the Earth, all these three come under the category of planets while the Moon is a natural satellite of the Earth. All the three revolve around the Sun and the Moon revolves around the Earth. So the Moon comes under a different category.

12. Which of the following is not a planet ?

(a) Moon

(b) Earth

(c) Mars

(d) Mercury

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

See the explanation of the above question.

13. Match the following -

(A) Planet

(1) Moon

(B) Satellite

(2) Uranus

(C) Comet

(3) Mariner

(D) Artificial spacecraft

(4) Halley

Code :

(a) A-2 B-1 C-4 D-3

(b) A-1 B-2 C-3 D-4

(c) A-4 B-3 C-1 D-2

(d) A-2 B-1 C-3 D-4

U.P.P.C.S. (Pre) 1992

Ans. (a)

The correct match is –

Planet - Uranus

Satellite - Moon

Comet - Halley

Artificial Spacecraft - Mariner

The Moon is the only satellite of the Earth. Uranus is a planet, which in terms of distance from the Sun is in 7th place, Mariner-9 is an artificial spacecraft launched to explore the planet Mars, on 30 May 1971 by NASA of the U.S.A. Halley is a comet that was seen from the Earth in 1835, 1910, and 1986.

14. The 'blue moon' phenomenon occurs when –

(a) Two full moons occur in the same month.

(b) Four full moons appear in two consecutive months of the same calendar year.

(c) Two full moons appear in the same month, thrice in a calendar

(d) None of the above

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Mains) 2007

Ans. (a)

A blue moon is a phenomenon involving the appearance of an additional full moon within a given period. Since a full moon occurs roughly every 29.5 days. After every 2.7154 years, an additional duration of 29.5 days accumulates in the calendar giving way to an additional full moon. It is called a blue moon.

15. When any object is taken from the Earth to the Moon –

(a) Its weight increases

(b) Its weight decrease

(c) There is no change in its weight

(d) It becomes totally weightless

U.P.P.C.S. (Pre) 2006

Ans. (b)

Weight of any object on any planet is equal to the product of its mass and gravitational acceleration. The Moon's gravitational acceleration is 1/6th of Earth. Therefore, when an object is taken to the Moon its weight decreases to 1/6th of its original weight.

Formula: Weight of Object = Mass × gravitational acceleration.

16. The distance of Moon from the Earth is –

(a) 364 thousand km

(b) 300 thousand km

(c) 446 thousand km

(d) 350 thousand km

U.P. Lower Sub. (Pre) 2008

Ans. (a)

The average distance of the Moon from the Earth is 384400 kilometers. In the given options only the option of 364000 kilometer is closest to the actual figure. The minimum distance between the Moon and the Earth is 363300 km and the maximum distance is 405500 km while the average distance is 384400 km.

17. A country by sending its 'SELENE' (Selenological and Engineering Explorer) in the orbit of the Moon became the first country in Asia. The country which achieved this fame is –